ActInf Textbook Group ~ Cohort 1 ~ Meeting 22 (Chapter 9, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 22 (Chapter 9, part 2) | 57:45

Hello, thanks for all joining. It's ACTIV textbook group, cohort one, meeting 22, chapter nine on October 28th.

Okay, we're in the second discussion on chapter nine.

Model-based data analysis.

Would anyone like to add anything they want or mention how the generative model discussion from the previous one hour relates to model-based data analysis?

How does chapter four relate to chapter nine?

Okay.

What does chapter nine address?

What do chapter six and nine address that chapter four doesn't?

I mean, six is about thinking about applying it in a, like, incredibly kind of simplified, step-by-step kind of way.

And nine is very specifically about answering scientific questions with it and doing that with a sort of...

sort of schema of active inference.

So, I mean, that's...

So, I mean, that's...

the through line, I guess?

But, um...

I'm not sure more than that what to say.

No, that's really insightful.

Ali?

Ali?

Yeah.

Actually, I would compare it to a kind of learning how to solve problems in, I don't know, mathematics or physics or other areas.

For example, in chapter four, we acquired some necessary knowledge or prerequisites about how to...

how to think about the phenomena or the problem we're dealing with.

And chapter six kind of outlines a roadmap, a roadmap about how exactly we should proceed to solve that problem in a kind of formulaic way.

And chapter nine, actually, in chapter nine, we get our hands dirty and try to solve some of the actual or empirical problems with the tools or techniques we had learned in chapter four and six.

So, in this case, I somehow think of chapter nine as a kind of case study.

Well, a kind of case of studies or a kind of solved problem section of this textbook.

And, yeah, I don't know how much that captures their nature.

That's great.

Wow.

Great thoughts.

I totally agree.

Let's look at the table of contents.

Chapter one, special overview chapter.

Mirror symmetry, same but different with ten.

So, now in the middle eight chapters.

Two chapters.

Two chapters on the low road and the high road.

Helping to two different approaches which can be developed like in a lot more detail to approach active inference from two fascinating and non-controversial starting points.

Bayesian inference from the low road, free energy principle, and the repeated measurement persistence imperative.

Chapter four is the first chapter on active inference.

Active inference is about the generative models that are being specified and how they're being computed.

After just one chapter on active inference.

It goes to the area where the most research and modeling has been done in active inference.

Which is computational man alien neuroanatomy.

One chapter on active inference.

One chapter on active inference.

One chapter on the primary domain.

The first half of the book.

The second half of the book.

The second half of the book.
The second half of the book.
The second half of the book.

The part where you're going to be in parallel, perhaps on your first, but perhaps not on your first reading, you're going to be playing through these applied aspects of modeling.

First is the recipe for designing active inference models in chapter 6, as both of you have very nicely said.

Then, the bigger 4.3 dialectic discrete continuous time is revisited in two chapters.

Because it probably, it's one key difference, it's a minimum of two, so it prevents any like, oh, well, it has to be a PMDP.

It's like, well, obviously not.

So it keeps the space of generative modeling open, and it allows for discussion of the very interesting topic of hybrid active inference nested models with continuous actuators, for example, but discrete decision-making apparatus.

So, it just is like wanting to cover these two key types of models in a bit more detail than was addressed in chapter 4.

But this was the first possible moment to address it after 4.3.

Just kidding.

Maybe not.

Then, chapter 9.

Or, yeah, where we are now.

Which is just, okay.

You've built your pure discrete time, pure continuous time, or hybrid generative model.

If it's a toy model, you can just generate data and just do whatever you want.

Or, you might have data from sensors, or real world, or however, that's in the data format that your generative model expected to receive.

Which was part of your design process.

You could have determined what structure, literally, the data is going to be in.

Is it going to be like, yes, data can be reshaped.

But just broadly, is it like 256 sensors in a row of one?

Or is it like 8 by 8?

Just those kinds of questions.

So that when the data start coming in for the experiment, whether you generated the data from the generative model, or whether you're using it in a purely recognition model capacity.

On real world data, what does it look like to do statistical modeling with these statistical models that we've been using?

And then, chapter 10 is a review.

It restates what each chapter does.

Provides some further thoughts.

And just summarizes the unifying capacity of active inference modeling.

Then there are the appendices.

And also one, in my opinion, very interesting section of chapter 9 is the section on models of false inference, section 9.7, in which there's an interesting definition of the disorders.

I mean, the mental disorders.

And how we can actually computationally model those different disorders.

Sorry, it says that here the disorder does not necessarily imply that the inferential mechanism is flawed.

In most of the studies, the inferential mechanism operates normally, but based on a flawed generative model.

So, that's basically what we had learned about how we can somehow update our beliefs or correct our perceptions based on either refining our generative model,

or modifying our generative model, or somehow, sometimes we can just, well, act upon those errors and try to reduce our uncertainties.

But in this case, it's a very interesting point on the case of computational pathologies.

Because oftentimes people ask, well, what does active inference actually explain?

So, I think this is a very good example of its very meaningful and significant application, I mean, in any sense of the word.

So, if we can somehow computationally model these kinds of disorders, we'll most probably have a much better understanding of its origin and the way to probable treatments of those disorders.

And many other applications can come to mind.

But, yeah, I think this part of the chapter 9 is something that was not addressed quite explicitly in the previous chapters.

And I, in my opinion, is the most interesting part of this one.

Great. Awesome.

Yeah, it reminds me of the table in chapter 4.

Or, sorry, chapter 5 with the neurotransmitters.

So, that was, they were still at the organ systems level in chapter 5.

But 5 is very much related to 9, I guess, in that sense as well.

Because it's like, it's like, here's what dopamine does, here's what acetylcholine does.

Just at the computational model level.

And then here it's like, that computational model was fit, like, between the phenomena, like attention, or policy precision, or however.

Phenomena to a molecule.

In chapter 5.

Here, the phenomena is being connected to a real human clinical application.

So, it's not that we're saying, or anyone is saying, or anything like that.

That the molecule is being directly linked to this paper.

Even if it's mentioned.

The direct linkage approach is, people in group X have more molecule.

Have less molecule.

And even the neuroimaging approach can be seen as a variant of that.

Which is, people in group X have more brain activation pattern.

In computational pathology, those kinds of biochemical measurements and neuroimaging measurements can and they are used.

However, this type of computational modeling is referring to parametric modeling where the parameters are related to the pathology.

So, it's not an a priori classification of those with and without.

And then descriptive statistics showing that a biochemical or a brain imaging pattern are differentially expressed.

That's descriptive statistics.

It's not computational pathology.

It might be precision pathology.

But it could still be precision or big data or however with just like purely doing data summaries.

Doing a random forest model to diagnose.

These models have a cognitive structural hypothesis about the nature of these phenomena.

Like the ocular motor syndromes.

Again, it's not just people with the diagnosis and without.

It's a neurocognitive model of eye behavior.

And then explanations can be made from clinical data.

By rendering various aspects of the generative model conditionally independent of others, ocular motor syndromes such as intranuclear optomolegias may be induced.

So, if you just were diagnosing these as different conditions, maybe they have similar etiologies.

How the condition arises?

Maybe they have different.

Maybe they have multiple.

But these two conditions computationally can be understood as parametric modulating in a computational model of ocular motor behavior.

This isn't people with internuclear opto-data have higher glucose or lower glucose or this brain region or this aspect of them.

Those are all absolutely valid approaches.

That's like scientific methodological pluralism.

Which is like, it's okay that people want to do different approaches.

Everybody's contributing to a common...

People who care about this and are creating high quality reproducible data, their data will always be useful.

And their perspectives are valid and we don't want to have a discourse space on internuclear optomolegia.

That's, you know, like saying that one model is like simply the only correct model.

So, this paper makes a positive contribution and points to some new ways of thinking about various pathologies.

And thinking beyond the ways that they've been worked on before.

Where one can imagine that before 2000 and so on, you had what?

Health records.

And you have some double-blind experiments.

And so, you might do a meta-analysis to find correlations with features of people and all of those things that we've been describing.

But it was not plausible to actually implement computational models of these phenomena.

Now, interestingly, that's something that can be worked on by researchers directly.

And then somebody can...

A doctor could download ocular motor syndromes dot...

Programming dot...

You know, whatever it is.

But this can be developed in a pure research context.

And the generative model is like the interface...

Or the kind of information...

That might not be available to the person developing the generative model.

Because it relates to a lot of problems and issues about data...

Availability.

That's one kind of just side note.

But that's...

But, Lily, thanks for raising that point.

That I just wanted to emphasize...

And unpack.

Which is that these...

This table, which is taking up two and a half pages of a pretty short book...

Is describing...

These are like...

The cases that can be brought up.

Which they're providing in a helpful table...

For what active inference describes.

So if the principle of least action...

Gauge theory, etc.

Sophistication...

Does not interest one...

These are more pragmatic value.

Now, that will probably move the goalpost to...

Show me where the computational modeling...

Helped somebody with delusions.

That's...

And that's a fine place to want to go.

But it's also a standard and an approach that...

It's not related to the substance of the theoretical development.

I mean, I think you could probably...

The response to that question might be something like...

The difficulty of...

Implementing that model...

In a way that provides...

You know, actionable...

Real-time...

You know...

Affordances to address...

The...

Aberrant priors...

In a way that's...

Causally...

You know...

It's going to actually affect...

That...

System...

Is...

We need...

You know...

Either more compute...

Or...

A particular kind of...

Algorithmic medicine...

That doesn't exist yet...

Like...

But that's not...

A problem with...

A particular approach...

It's just an...

Insufficiency of...

Ability to...

Enact it...

I guess...

But I wanted to ask a question...

About...

Specifically this...

Aberrant...

Prior beliefs...

It says...

Ultimately the pathology...

Is a consequence of...

Advert prior beliefs...

It's not the generative model...

It's the average prior beliefs...

Is that...

Um...

I think...

I think this is...

Like...

Ellie's point is...

Everything that you just...

Went through is correct...

That this is kind of...

Where the rubber really meets the road for...

Its...

Value in demonstrating...

Or whatever...

Um...

But...

Um...

I'm wondering also...

To me that...

There's like a...

Easy confusion...

Thing...

There about...

Beliefs...

And priors...

And so...

Um...

Well if they just believed the right thing...

Then they wouldn't have these...

You know...

Um...

That's not really what that means...

Though right...

So...

Like...

Is another way of saying this...

Like in the ocular motor...

Uh...

You know...

Disorder...

Case...

Like a...

You know...

Kind of...

Uh...

Neurobiological...

Like...

You know...

Bias...

In...

In some kinds of...

Cells...

Or whatever...

Right...

For...

That's like...

You know...

Normal...

Or whatever...

Like...

Some...

You know...

Receptor...

For a particular...

Chemical...

Or whatever...

Is that...

A kind of thing...

That...

A way of saying that...

Differently...

That...

In that particular case...

That is...

More commensurate with what they're actually saying there...

Or like...

Not more commensurate...

But just...

Not...

Relying on...

Relying on...

Jargon...

And or...

Domain...

Knowledge...

Specific...

For what they mean by beliefs and priors...

Is that...

Or...

Is that not right?

Yeah...

Well...

There's a lot...

There...

Are you...

Talking about...

The...

Comparison of the...

Terms...

Beliefs...

And priors...

In relationship to the way people might...

Naturally...

In relationship to the way people might...

Naturally interpret...

Such...

Beliefs to be...

And they may indeed be...

People's beliefs about the world...

In the conversational sense...

Like some beliefs are...

Yeah...

Metinal states...

Yes...

And those...

But other states of the world...

As if...

Or really...

Are your states on other things...

Right...

Yeah...

I guess I'm asking if...

Beliefs...

Is...

In some cases it is beliefs...

It's also...

Another...

Way of stating...

If...

Another way of stating...

That is...

Like a...

Like...

Physical...

Like a structural...

Like systemic bias...

For one thing...

Or another...

Or whatever...

In that context...

Is that...

You know...

The...

Like...

Physical...

Or informational...

Non...

Folk psychology...

Is that...

Commensurate with what they mean by...

Beliefs...

There?

Okay...

Let's try to...

Yeah...

Ale...

Sorry...

Uh...

I'm sorry...

But...

You mean...

The...

Statistical use of the word...

Belief...

Because...

In...

Statistics...

And...

In folk psychology...

As far as I know...

This term...

Is used in quite different...

Ways...

Yeah...

Uh...

I...

Guess that...

What...

What you're talking about...

Is...

Uh...

The statistical sense of the word...

Belief...

And...

If it...

Maps onto...

Uh...

The folk psychological...

Uh...

Sense of this...

Term...

Or...

Uh...

Am I missing something...

Here?

Um...

Well...

Let's just take two steps back for a second...

And just...

In the text it says...

Um...

At the bottom of page 185 here...

The...

At the top of that...

Yeah...

It says...

Um...

Um...

The inferential...

Uh...

Mechanism operates normally...

But based...

On a flawed generative model...

This means that ultimately...

Pathology is a consequence of aberrant prior beliefs...

!

Yeah...

And I'm...

And I'm asking...

Um...

A kind of...

Just to...

Explicate different...

Whatever...

Uh...

Right there...

In...

In the context of psychological disorders...

Which is not necessarily pathology...

You know...

If you tell somebody...

It helps with...

Uh...

Or is the...

The impetus...

Or whatever...

For this whole thing was...

To think about...

Um...

Yeah...

Like...

Schizophrenia...
And other...
You know...
But that's...
Immediately...
Kind of...
Puts it in that category...
Or that context of...
Definitions...
And so...
Uh...
Just to...
Um...
Kind of...
You know...
Nuance the term there...
Beliefs...
A little bit...
Um...
Yeah...
Is it commensurate with the rest of the text?
Uh...
To say...
Like...
In the case of this...
You know...
Like...
Specifically...
Of...
Uh...
Ocular...
Uh...
Disorder category...

You're saying...
Beliefs in the retina...
Or something...
Like...

That is...

Um...

Like...

Cones...

Like...

Being sensitive to something...

They...

They...

Well...

It's just what they are sensitive to...

They're just sensitive to...

Or whatever the...

Relevant...

Structure is there...

But it's...

Sensitive...

To...

Something...

And that's...

It's belief...

You know...

In some sense...

Um...

I'm not looking for...

Yeah...

What's the...

Statistical definition...

Or anything like that...

But just...

Is that...

Is that completely off track...

Or is that...

I have a thought...

If you want to make a response...

Please...

Uh...

Yeah...

I think that...

Uh...

This term belief...

And...

It's...

Different meanings...

And different contexts...

Actually...

Uh...

Sort of...

Bleed into...

Each other...

And...

Uh...

There cannot possibly be...

Uh...

Separated...

Uh...

Cleanly from...

Uh...

From each other...

So...

In this case...

I believe that...

The word belief...

Uh...

In...

Is...

Partly...

Is...

Um...

This...

Folk psychological...

Uh...

Sense of the word...

And...

Partly...

It's...

Based on the...

Uh...

Kind of...

Philosophical...

Slash...

Epistemological...

Uh...

Sense of this word...

Because...

Uh...

Especially in epistemology...

Uh...

When we talk about belief...

Usually...

They don't mean...

Just the...

Uh...

Psychological...

Sense of this term...

Although...

Uh...

It relates to that...

Concept...

But...

It's much more nuanced than that...

So...

Uh...

In the case of...

Oculomotor...

Uh...

Pathology...

Well...

Uh...

I believe that...

Uh...

The word belief...

Uh...

Could be interpreted...

In...

Uh...

A little more abstract...

Uh...

Philosophical sense of...

Or...

Epistemological sense of this...

Uh...

Uh...

Than our...

Um...

Folk psychological sense of it...

So...

Uh...

But...

Uh...

Again...

Uh...

I think...

Uh...

It has some overlappings with...

Uh...

The folk psychological sense of it as well...

Great...

Thought...

Just to kind of...

Restate that...

Then...

Say what I was planning to before...

So...

Here...

It's situational...

Um...

Where people talk commonly about...

Uh...

Don't talk about beliefs...

Like the eye...

Like where it believes it's...

Should be looking...

Um...

People don't talk about that...

But...

There might be another cognitive model that does...

Align very well...

With an expressed...

Felt...

Cognitive belief...

And...

Everything in between...

And...

Bundles...

That are mixed...

Like people who believe that they're accurate on a task...

But actually are low accuracy...

So it's gonna be very model specific...

I think...

How it...

Maps to different...

People's understanding...

But...

What I was...

Uh...

That was what Ali's response...

Kind of made me...

Think about...

And that's why I didn't...

Choose an example like delusions...

To focus on...

Because that is related to...

Or experienced...

Or reported...

Phenomena...

Um...

Okay...

So...

Leave some priors...

And this question of disorders of inference...

Which as Brock mentioned...

Was like...

Part of the early impetus...

In relating to studying...

For example...

Schizophrenia...

And...

Fristin's earlier...

And continued work...

Maybe...

Maybe...

There's a...

Part of missing...

And maybe it's not the right...

Context or way...

But...

I think...

First...

We can...

Narrally...

Interpret...

What they use to mean...

Disorder...

And then have a more...

Broad perspective...

So...

I...

Read this...
Does not necessarily imply...
The inferential mechanisms flawed...
Like...
The CPU...
Has the right number of cycles...
The machinery...
Are not flawed...
The car...
Is operating at normal...
Burning temperature...
Given...
Its generative model...
As specified...
Hashtag chapter 4...
It is doing...
Appropriate...
Variational message passing...
Belief propagation...
Approximate...
Bayesian computation...
Bayesian brain...
However you want to see it...
Given what this person...

And their...
Parameterization...
It's not the case...
That they're making...
A fallacious estimate...
Now that's sort of...

Okay...
But you're going to say that...
For whatever they do...
But it's like...

That's kind of that...

Tautology...

It's like...

Right...

That's our best estimate...

Is just...

How it is...

In most of the studies...

Reviewed in table 9.1...

Which is to say...

The pathologies...

That are identified...

In table 9.1...

The inferential mechanism...

Operates normally...

The previous meaning...

But based upon...

A flawed generative model...

And it's literally like...

There's the rub...

Aberrant prior beliefs...

Is like...

And so...

That is where...

I would say...

There's two...

Ways to zoom out from that...

And contextualize that statement...

The first is...

Aberrant prior beliefs...

Are not intrinsically aberrant...

Whether they're...

Subconsciously...

Or consciously held...

Belief in itself...

Is not aberrant...

But...

There's the...

Niche...

And especially...

Just the human cultural...

When we're talking about...

These felt beliefs...

In the niche...

It's like...

There is...

Ant colonies...

That were high foraging...

And low foraging...

Those are not...

Aberrances...

Of foraging behavior...

That's...

Variation in foraging behavior...

With respect to...

Variation in weather...

And there was probably...

Decades where...

One was better than the other...

And that's why...

There's evolution...

So it's like...

What's the...

Okay...

So...

Level one of aberrant...

Is in a niche context...

And then...

Especially for these...

Like...

Expressively...

Human...

Experienced...

Settings...

That's where I think...

There's an opportunity...

Again...

Not saying that this paragraph...

Is the right place...

In this textbook...

But that's where...

It's important to also...

Bring in like...

Like...

I'm not...

Expert in this area...

But like...

Foucault...

Social...

Descriptions...

Of...

Mental illness...

Diagnosis...

People are only crazy...

If...

The majority of people...

Yeah...

Crazy...

Something...

This...

Book...

Yeah...

Which I've mentioned...

On live streams before...

If I could just...

Describe why...

I bring it up...

And like...

How I think it's relevant...

To chapter nine...

Is just like...

She's saying...

There's different methods...

And there's different...

Deoretical interpretations...

And it's very...

Clear...

Pluralist...

And...

Though guided...

Everybody will also...

I think...

Deeper...

Aspects too...

But just...

I...

Studied with Helen...

In...

Stanford...

And she...

It was just very...

Like...

It was very interesting to...

See that...

Coming from philosophy...

Because...

In the scientific domains...

It's like...

Not framed that way...

It's like...

We could have a generative model...

For...

These...

Conditions...

And...

Scenarios...

That integrates...

Different...

Types of information...

In a situational way...

With a sparse model...

And we could all...

Work on that...

And...

Then...

Different people...

Could have different...

Data sets...

Or different...

But...

That...

Requires...

Not just...

Operational coordination...

And...

Ontology driven...

Design...

And other features...

But it also requires...

A...

Basically...

Personally held...

Commitment to pluralism...

At least...

In practice...

Because...

There's too many methods...

And methods are too...

Last mile...

To...

Models of phenomena...

These are not measurements...

But...

Alongside...

And with...

And so...

That's not...

Descriptive statistics...

Per se...

That's a major difference...

That's a major difference...

That's a major difference...

That's a major difference...

That's a major difference...

That's a major difference...

side and with and from measurements.

And so that's not descriptive statistics
per se.
And that's a major difference in this, what you can do with the tail of two
densities
approaches these situations
differently than one might see
in a health journal showing
the population level
factors associated with a disease.
This is a different kind of study.
But it could be done in a population context
which has been done extensively in SPM and neuroimaging.
So you can still bring in features
of the people and understand how different features of people are quite literally associated
with the diagnosis.
And that is seen as a label on a neurocognitive model, a generative model.
Now, what's the generative model?
Chapter 4, chapter 6.
And then it's like, okay, we're just going to say that all of the people's ocular motor circuitry
are structurally the same.
And we're going to explore how the two groups differ in the parameter estimates of these different features.
Or you could say,
or you could say, I'm studying a situation where this region has been severed.
So the hypothesis I'm testing is actually whether this connectivity
versus this connectivity
is going to have this observable difference.
That's testing a structural hypothesis.
So you could test parametric hypotheses within
a generative model
and or structural hypotheses across.
But to kind of return to this point,
those are these two ways
that the model can be, quote,
in their way,
aberrant or flawed.
Quite literally,
it could be a parametric estimate
that's aberrant,

which, as discussed, does not mean intrinsically bad.
Number 4.2 or 3.7.3 is not bad.
It's about its relationship within a niche and fitness context
and like all of that.
So it could be a parametric aberrance
with respect to a state estimator,
like an expectation on a varial,
variable or precision.
So that's the Laplace approximation.
It's the hierarchical predictive processing approximation
slash structure.
Expectation, variance.
Mean, variance.
Mode, variance.
Center of the path,
width of the path.
Those are the Gaussian distribution.
Takes two parameters.
Laplace approximation takes two parameters.
Hierarchical message passing
has these kinds of variables.
Or it can be structural differences,
which is quite open-ended.
How is this useful?
An example is provided here.
Potentially, for example,
let's just say that there's people
who do and do not have a certain situation.
Some moths go towards the light
and some don't.
Is that because of differences
in their behavior
where the statistical variance
gets loaded onto differences
in their realized preference vector
or their habit, E?
So there could be a behavioral setup
that would do, you know,

the Fibonacci pattern of variability
or just whatever it happened to be,
which is extensively described in SPM,
how to time different behavioral stimuli.
Some natural observational setting
or controlled behavioral setting,
both of which have pros and cons,
both, quote,
controlled conditions in the lab
and natural observation.
They have pros and cons.
But the idea in model-based analysis
is that those can be the observables
and you may or may not control
the observations,
but at least you know what they are.
And then you can explore
whether differences in behavior
end up being reflected
in statistically differentiable
parameter estimates
for prior preference
or for prior D
or for the fixed form term E.
Ali.
And yes, for what it's worth,
I also put a quote
from this recent paper
by Carl Pristen
from a couple of months ago,
which is, by the way,
is a great overview
of this whole field of computation
psychiatry
through the lens of active inference.
And here it provides
a clear and explicit definition
of the term belief

used in this area.
So I thought that might be helpful here.
The kind of belief,
beliefs here are Bayesian beliefs
and then it goes on to describe
what Bayesian beliefs mean.
Awesome.
Cool.
That was super clarifying,
I think.
It makes a lot of sense.
And I have still a kind of,
maybe,
I don't know if
the answer
was already given exactly
or I'll just ask it
a different,
slightly different way
about
the separation
between the generative model
and the priors here
in
this toy,
you know,
frog jumping example
are
are the
avarant priors
the,
you know,
the frog apple
distribution
on that belief
beforehand
and
the observation,

the updating,
all of that
being the generative model
is working as,
you know,
intended,
but it's just that
the
priors
were
so out of whack
that when,
you know,
the
generative model
kind of observes
and steps through
or whatever
that it's,
you know,
it just ends up
in the wrong state
and or
is the likelihood
model.
It just,
I'm not sure,
I guess,
in this,
in the toy example
there,
how you could have,
how it would be,
I guess,
the part that's missing
there is the context
maybe,
but,

yeah,
like if I had
a prior belief
that was different
than your prior belief,
but we weren't,
you know,
we didn't have a disorder
or whatever,
I would imagine
our priors
wouldn't be the same,
but that would still
kind of converge.
So I'm not sure
how you get there
without
an outside prior belief
or something
about the environment
that's totally wrong,
right?
Or,
like,
or,
you know,
something else
about the environment
or the likelihood model
is somehow
really wrong?
Yeah,
thanks.
Great question.
Those are exactly
the two.
Greetings,
welcome,

faculty.
Those are exactly
like the two.
Let me unpack that
in the pathology example.
So,
society expects us
and or
we actually
die
or have consequences
with
when the frog,
when it's a frog,
you're not supposed
to eat it.
Apples
are okay to eat.
Now,
if you
turn down
too many apples,
you starve.
If you
eat too many frogs,
you die.
So,
you have to
balance your
type 1
and type 2
false positive
and false negative
error rates
in this
eat,
do not eat,
jump,

does not jump
observation
setting.
because these
are idle
classifiers here.
But we can
imagine these
as more
action-oriented
classifiers.
Apples are
for eating,
frogs are
for throwing.
now,
society and
or the
niche
defines
aberrant
behavior
as
one that
does not
conform
to
expectations
or
to the
to the
to a
thriving state
in the
niche.
While
recognizing
that all

parameter values
that are
actually observed
are at least
there.
now,
that doesn't
mean you
want them
to be there.
They want
them,
but also
then that's
a belief
an agent
has about
another
system
that also
exists.
So,
there would
be many
ways
to see
that
aberrant
behavior.
So,
if you,
like,
speaking to
the person
who discards
too many
apples,
maybe they

believe that
they have a
strong prior
on frogs.
0.5
or their
likelihood,
which is also
a parametric
belief,
is different.
They have a
different action
mapping.
Now,
here's the
question.
Now,
you put them
in the
lab and
you have
jumping and
non-jumping
frogs and
apples.
And then
you can use
those
experimental
techniques
to,
on a
computational
model,
narrowly
to determine
do they

simply
expect
frogs to
appear,
but they
know what
frogs and
apples do,
or
do they
have an
appropriate
belief about
the likelihood
a priori of
frogs and
apples,
but they
have an
incorrect
action
mapping.
frogs.
I hope I
said it
correctly,
but I
think you
get the
picture.
The point
is you
could then
use
experiments
to determine
whether
groups of

people who
had
effective
or
ineffective
behavior,
and then
if somebody,
for example,
had an
overestimate
of frogs,
then the
message to
pass is
frogs are
not as
common as
you think.
Whereas if
there was
somebody with
a variance,
interpretable
variance,
here in the
likelihood model,
difference with
some other
preference,
again,
keeping in
mind that
that's still
like,
it's like
outlier
detection

and
attracting,
which isn't
even necessarily
a preferable
thing,
but just
pointing out
just statistically.
This person's
model would be
updated by
saying,
you know,
it's not
known that
apples can
jump,
but actually
they can.
It's possible
that apples
jump,
so next
time something
jumps,
don't just
throw it
away.
And those
are very
different
points to
convey.
And this
is a very
simple
example,

but it's
helpful to
think about
because those
are the
kinds of
parameters.
If this
was your
model of
schizophrenia,
those would
be your
levers or
your knobs.
If you had
a more
complex model
and the
models are
more complex,
then there's
a lot more
associated
challenge.
Yeah,
Brock?
that was
really helpful.
And I
don't even
know.
It seems
good.
I mean,
obviously,
I know next
to nothing,

I guess,
about
psychiatry.
I don't
know how
you would
do it
without
something
like
this.
I mean,
I guess
they were
doing something
kind of like
this approximately
informally,
to some degree
in some cases,
but it would
seem like
just guessing
almost.
I don't
know.
Well,
I could
infer about
schizophrenia
without something
like this.
Well,
to simplify
a complex
area,
there was
studies that

associated
either a
biomolecule
or a genetic
variant,
but especially
biomolecules
and brain
imaging,
directly to
a diagnosis,
which from a
health record
perspective,
exactly makes
sense.
It's the data
you have.
However,
this is
an unobserved
model of
the phenomena
itself that
can incorporate
measurements,
including health
records,
and that
allows the
development of
an actual
understanding of
the etiology
and the
causation,
and therefore
could be used

to justify
traditionally
understood
interventions
or suggest
different
behavior
and
interventions.
Like,
maybe you
don't,
oh,
you're low
on this,
just take
this thing
that raises
it.
It's like,
maybe that's
not exactly
the move.
more precise
affordance to
do that.
Yeah,
including the
counterfactuals,
like in a
clinical setting,
that is going to
be quite
interesting.
All right,
let's just,
in the last
minutes,

look to
10.
Minsky,
active
inference as a
unified theory
of sentient
behavior.
If anybody
wants to do
a little,
like,
Twitter thread
recap on
the sentient,
the sentience
discourse over
the last few
weeks,
I'm sure we
can find some
funny tweets.
But with the
sentient pong
playing neurons
and just
wave after
wave of
semantic
discussion
about the
term
sentient,
in this
chapter,
we wrap up
active
inference main

theoretical
points from
the first
part of the
book,
chapter 1
through 5,
and its
practical
implementations
from the
second part,
6 through
10.
Then we
connect the
dots.
We abstract
away from
the specific
active inference
models discussed
in previous
chapters to
focus on
integrative
aspects of
the framework.
Benefits
of ActInf.
Summary of
the book,
10.2,
chapter 1.
Chapter 2.
Chapter 3.
Chapter 4.
5.

6.
7 and 8.
No love.
Chapter 9.
You just
read it.
Go read it
again.
Connecting the
dots.
The integrative
perspective on
active inference.
Interestingly
framed in
terms of some
early
Dennett
work.
a call to
model the
whole iguana,
a complete
cognitive
creature,
perhaps a
simple one,
and an
environmental
niche for
it to
cope with,
rather than
single-dimensional
!
It's a

thing.

It's a

It's a
simple
thing.
It's a
simple
thing.
season measurements
on a complex
system.
It's just
too much like
a whirlwind
to just
measure and
brain regions
correlated with
this, and
this region is
correlated with
that.
those kinds
of structural
and
behavioral
morphological
any given
measurement
does not
ever get
at the
underlying
causal
structure,
unless you
specifically

model it
that way.
Active
inference helps
with that
issue by
offering a
first-principal
account of the
ways in which
organisms solve
their adaptive
problems.
So by
framing
attention,
memory,
anticipation,
under
common
computational
framings,
and a
unified
imperative,
it's
possible to
integrate
different kinds
of phenomena.
Memory,
attention,
can be
thought of as
an optimizing
same objective.
10.4.
Predictive

brains,
predictive
minds,
and
predictive
processing.
Kind of a
provocative
selection.
Active
inference
creatures or
their brains
are probabilistic
inference
machines.
Connecting
to the
kind of
focus of
predictive
mind,
predictive
brain,
predictive
processing,
coding
area,
which is
like the
real-time
and
anticipatory,
especially
real-time
unfolding,
anticipation,
self-evidencing,

prediction.
Predictive
processing
specifically,
Livestream
43.
Perception,
section
10.5.
Helmholtz,
perception
as
unconscious
inference.
Then,
how that
was implemented
in
Bayesian
brain.
Bayesian
brain.
The inactive
turn,
the pragmatic
turn,
action
control.
Bringing
action in.
End of
Livestream
43.
End of
paper
43.
idea
motor,

cybernetics,
some other
ontologies that
are kind
of like
easily
functionally
equated with
parts of
fact-inf.
Idea
motor theory,
cybernetics,
optimal control
theory,
utility
decision-making,
Bayesian
decision,
reinforcement
learning,
planning as
inference,
behavior
and bounded
rationality,
and then
specifically
what free
energy brings
into bounded
rationality,
valence
emotion
and motivation,
homeostasis,
allostasis,
interoceptive

processing,
attention
salience,
epistemic
dynamics,
rural
learning,
causal
inference,
fast
generalization,
and next
steps.
Social,
machine
learning and
robotics,
summary,
Lord of the
Rings,
we are
confident that
you will
continue to
pursue active
inference in
some form.
So it's a
quite long
chapter,
there's a
lot in
it.
We'll talk
about it
next time.
Thank you
fellows.

Bye.